

CLINICAL COMMENTARY

MIDFOOT AND FOREFOOT INVOLVEMENT IN LATERAL ANKLE SPRAINS AND CHRONIC ANKLE INSTABILITY.
PART 1: ANATOMY AND BIOMECHANICSJohn J. Fraser, PT, MS, OCS^{1,2}Mark A. Feger, Ph.D., ATC³Jay Hertel, Ph.D., ATC¹

ABSTRACT

The modern human foot is the culmination of more than five million years of evolution. The ankle-foot complex absorbs forces during loading, accommodates uneven surfaces, and acts as a lever for efficient propulsion. The ankle-foot complex has six independent functional segments that should be understood for proper assessment and treatment of foot and ankle injuries: the shank, rearfoot, midfoot, lateral forefoot, and the medial forefoot. The compliance of the individual segments of the foot is dependent on velocity, task, and active and passive coupling mechanisms within each of the foot segments. It is also important to understand the passive, active, and neural subsystems that are functionally intertwined to provide structure and control to the multisegmented foot. The purpose of the first part of this clinical commentary and current concepts review was to examine foot and ankle anatomy, detail the roles of the intrinsic and extrinsic foot and ankle musculature from a multisegmented foot perspective, and discuss the biomechanics of the ankle-foot complex during function. The interplay of segmental joint mobility, afferent and efferent sensorimotor function, and movement and stabilization provided by the extrinsic and intrinsic musculature is required to coordinate and execute the complex kinematic movements in the ankle-foot complex during propulsion.

Key Words: intrinsic foot muscles, gait, joint mobility, kinematics, ambulation

Level of Evidence: 5

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BACKGROUND AND PURPOSE

The modern human foot is the culmination of more than 5-million years of evolution.¹ Our ancient hominine ancestors evolved from arboreal to terrestrial living and the morphology and function of the foot adapted accordingly and transitioned from primarily climbing tasks to bipedal locomotion.¹⁻⁵ The ankle-foot complex absorbs forces during loading, accommodates uneven surfaces, and acts as a lever for efficient propulsion.⁶ The six independent functional segments that comprise the ankle-foot complex should be understood for proper assessment and treatment of foot and ankle injuries: the shank, rearfoot, midfoot, lateral forefoot, and the medial forefoot (first ray and hallux).⁷ Compliance of the individual segments of the foot is dependent on velocity, task, and active and passive coupling mechanisms within each of the foot segments. It is also important to understand the passive, active, and neural subsystems that are functionally intertwined to provide structure and control to the multisegmented foot.^{8,9}

The purpose of the first part of this clinical commentary and current concepts review was to examine foot and ankle anatomy, detail the roles of the intrinsic and extrinsic foot and ankle musculature from a multisegmented foot perspective, and discuss the biomechanics of the ankle-foot complex during function. The companion paper to this commentary will examine the contribution of midfoot and forefoot impairment in lateral ankle sprains and chronic ankle instability in order to increase clinician's awareness and to facilitate future research in this area.¹⁰ The importance of multisegmented foot and ankle assessment will also be discussed from a clinical and research perspective.

ANATOMY AND FUNCTION OF THE ANKLE – FOOT COMPLEX

Shank

The shank (tibia and fibula) begins at the knee where the tibia articulates with the distal femur and the fibular head articulates with the lateral tibial condyle. The tibia and fibula are supported by the anterior and posterior ligaments of the fibula head proximally, by the interosseous ligament along the diaphysis, and the anterior and posterior tibiofibular

ligaments distally. The shank forms a distal mortise joint that serves as the proximal segment of the talocrural articulation. During gait, the shank functions like the distal aspect of a pendulum during swing phase and fulcrums over the talus during periods of single support.¹¹ In stance, the fibula shares approximately 10% to 30% of the burden during axial loading of the shank.¹² The shank is coupled to ankle motion with tibial internal rotation coupled to rearfoot pronation^{13,14} and talocrural dorsiflexion/plantarflexion coupled to fibular translation and rotation in all cardinal planes.¹⁵

Talocrural Articulation

The talocrural articulation is a “mortise and tenon” joint comprised of the shank proximally and the talus distally. Dorsiflexion and plantarflexion are the primary osteokinematic motions of the talocrural joint, with an oblique axis of rotation that travels through the medial malleolus, the talar head, and the lateral malleolus.¹⁶⁻¹⁸ The oblique axis of rotation results in component eversion and adduction accompanying dorsiflexion and inversion and abduction accompanying plantarflexion.^{16,18} The talocrural joint is statically supported by the joint capsule, the deltoid ligament (medially), and the anterior talofibular, calcaneofibular, and posterior talofibular ligaments (laterally).

Rearfoot and Subtalar Joint

The rearfoot, also known as the hindfoot, is comprised of the talus (proximally), the subtalar joint, and the calcaneus (distally). The subtalar joint is comprised of anterior (talocalcaneonavicular) and posterior (talocalcaneal) articulations that are separated by the tarsal canal. The joint capsules and the cervical, interosseous talocalcaneal, posterior talocalcaneal, lateral talocalcaneal, calcaneofibular, and fibular-talocalcaneal ligaments support the subtalar joint. The axis of rotation is oriented anterior-superomedial to posterior-inferolateral and transects the three cardinal planes. Primary osteokinematic motions of the subtalar joint are supination and pronation. Similar to what is observed with the oblique axis of rotation in the talocrural joint, subtalar pronation is accompanied by dorsiflexion and abduction and subtalar inversion is accompanied by plantarflexion and adduction.^{16,18,19} The subtalar joint is

often described as a “mitered hinge” and its triplanar orientation allows for the coupling of shank rotation and rearfoot supination/pronation.

Transverse Tarsal (Chopart or Midtarsal)

Articulation

The transverse tarsal articulation is comprised of the talus and navicular (medially) and the calcaneus and cuboid (laterally). The bifurcate (calcaneocuboid and calcaneonavicular ligaments), dorsal calcaneocuboid, dorsal talonavicular, interosseous talocalcaneal, deltoid (tibionavicular part), spring, and plantar cuboideonavicular ligaments statically support the transverse tarsal joint.

Motion between the forefoot and rearfoot occurs in the transverse tarsal joint through two separate axes of rotation; one being supination and pronation of the cuboid on the calcaneus about the longitudinal axis and the other being oblique to the foot as the cuboid translates on the calcaneus.²⁰ The axes vary by task and change based on the congruency of the calcaneocuboid joint.²¹ The longitudinal axis of the transverse tarsal joint allows for the forefoot to rotate opposite of the midfoot in the transverse plane.²⁰ Arthrokinematic movement about the oblique axis plus dorsiflexion and plantarflexion of the transverse tarsal produces deformation of the longitudinal arches.²⁰ The review conducted by Tweed and colleagues²² is recommended for additional information on the function of the transverse tarsal joint.

Midfoot

The midfoot is comprised of the navicular, medial, intermediate and lateral cuneiforms, and the cuboid. The bones of the midfoot form the medial longitudinal (Figure 1), the lateral longitudinal, (Figure 2) and the transverse arches (Figure 3), which together comprise a half dome.²³ The function of the midfoot is to transmit and attenuate force and allow the foot to accommodate to the variable surface of the ground.²³ The arches are supported and controlled by a combination of bony congruency and static and dynamic extrinsic and intrinsic stabilizers.^{24,25} The primary stabilizers of the midfoot are evolutionary adaptations favorable for bipedal locomotion and consist of the osseous, muscular, and ligamentous structures that support the longitudinal

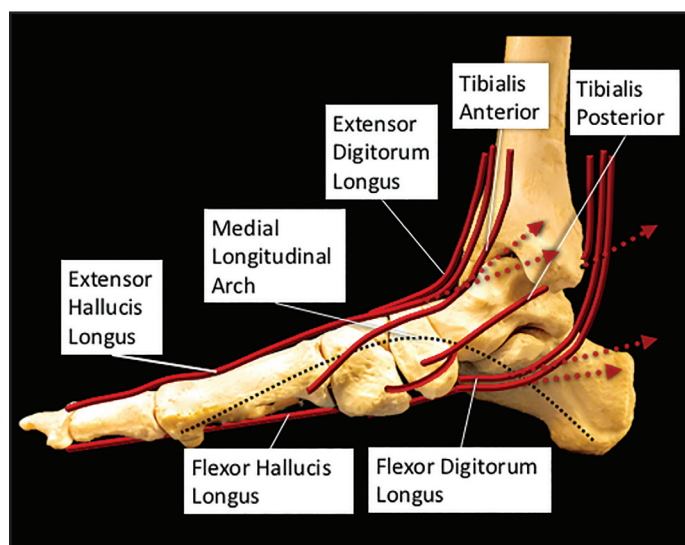


Figure 1. Morphology and Extrinsic Dynamic Support of the Medial Longitudinal Arch

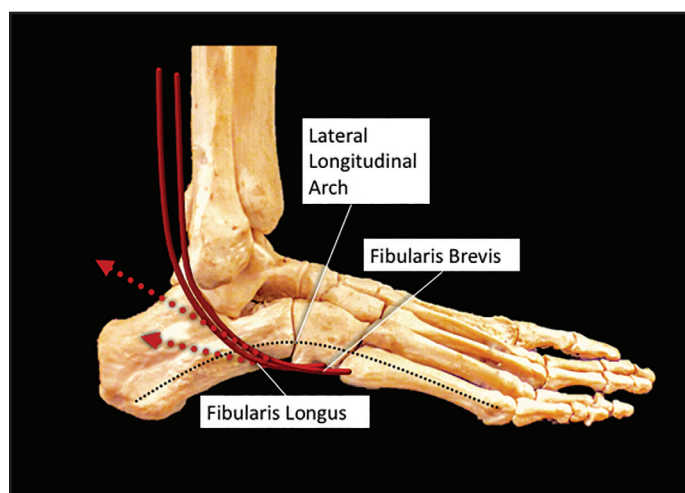


Figure 2. Morphology and Extrinsic Dynamic Support of the Lateral Longitudinal Arch

arches, calcaneocuboid, and tarsometatarsal articulations.²⁻⁵ The adaptations found in the human foot allow for various degrees of mechanical coupling of the forefoot to the rearfoot during gait (midtarsal locking) and control of dorsiflexion of the lateral tarsometatarsal joints (midtarsal break) during the stance phase.^{4,5,26,27} Table 1 provides a comprehensive overview of the stance phase of gait and details the multisegmented ankle-foot motions and the relevant muscle actions that occur from initial contact to pre-swing. During lower velocity locomotion, the foot is lengthened and the medial longitudinal arch is flattened, increasing the compliance of the foot

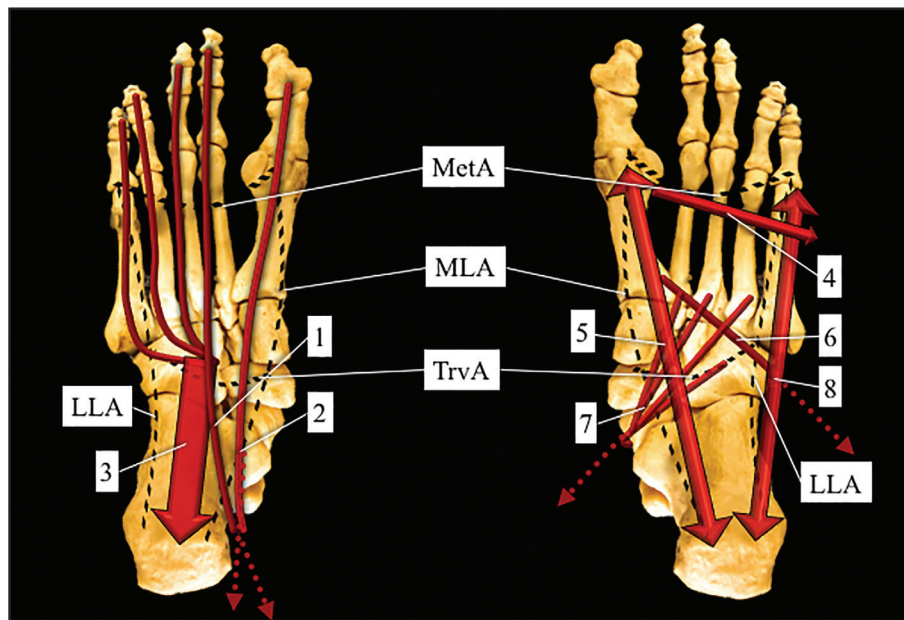


Figure 3. Morphology and Dynamic Support of the Longitudinal and Transverse Arches (Plantar Aspect). 1 = Flexor Digitorum Longus, 2 = Flexor Hallucis Longus, 3 = Quadratus Plantae, 4 = Adductor Hallucis, 5 = Abductor Hallucis, 6 = Fibularis Longus, 7 = Tibialis Posterior, 8 = Abductor Digiti Minimi, MLA = Medial Longitudinal Arches, MetA = Metatarsal Arches, LLA = Lateral Longitudinal Arches, TrvA = Transverse Arch

Table 1. Summary of multisegmented foot and ankle kinematics and muscle actions during stance phases of gait. L = Lateral, M = Medial, FHL = Flexor Hallucis Longus, FDL = Flexor Digitorum Longus.

Initial Contact			Loading		Midstance		Terminal Stance (heel off)		Pre-swing (toe off)	
Goal of Phase	Begin lowering forefoot to ground while maintaining proper frontal plane alignment		Dissipate ground reaction forces while accommodating terrain		Transfer momentum over foot and ankle complex with efficient transfer from deceleration to acceleration		Accelerate mass for forward propulsion		Finish forward propulsion into initial swing	
Relevant Multi-segmented Ankle-Foot Kinematics	Rearfoot:	Neutral Sagittal Pronated Abducted	Rearfoot:	Plantarflexed Pronated Adducted	Rearfoot:	Dorsiflexed Pronated Abducted	Rearfoot:	Plantarflexed Supinated	Rearfoot:	Plantarflexed Supinated
	Midfoot:	Supinated	Midfoot:	Dorsiflexed Pronated Adducted	Midfoot:	Neutral Sagittal Pronated Abducted	Midfoot:	Plantarflexed Supinated Abducted	Midfoot:	Plantarflexed Supinated Abducted
	L. Forefoot:	Plantarflexed Supinated Abducted	L. Forefoot:	Plantarflexed Pronated Abducted	L. Forefoot:	Dorsiflexed Supinated	L. Forefoot:	Dorsiflexed Supinated	L. Forefoot:	Plantarflexed Supinated
	M. Forefoot:	Dorsiflexed Everted Adducted	M. Forefoot:	Dorsiflexed Everted	M. Forefoot:	Dorsiflexed Everted	M. Forefoot:	Dorsiflexed Inverted Adducted	M. Forefoot:	Dorsiflexed Inverted Adducted
Relevant Ankle-Foot Muscle Actions	Tibialis Anterior: Maintains frontal plane alignment and eccentrically dorsiflexes foot		Tibialis Anterior: Maintains frontal plane alignment and eccentrically dorsiflexes foot		Fibularis Longus: Plantarflexes 1 st ray		Fibularis Longus: Isometrically stabilizes longitudinal and transverse arches		Fibularis Longus: Maintains 1 st ray on ground during propulsion	
	Fibularis Longus: Maintains frontal plane alignment prior to initial contact		FHL/FDL: Isometrically stabilizes longitudinal arches		FHL/FDL: Isometrically stabilizes longitudinal arches		Triceps Surae: Generates plantarflexion propulsion force		Triceps Surae: Generates plantarflexion propulsion force	
	Tibialis Posterior: Stabilizes rearfoot		Tibialis Posterior: Isometrically stabilizes medial longitudinal arch		Plantar Intrinsics: Accommodates terrain, stabilizes midfoot, and attenuates force					
			Triceps Surae: Tenses plantar aponeurosis							
			Plantar Intrinsics: Accommodates terrain, stabilizes midfoot, and attenuates force							

for accommodation of uneven terrain and to maximize balance control.²⁸ During higher velocity locomotion, the medial longitudinal arch angle increases and the foot shortens as a means of optimizing the lever arm during pushoff.²⁸

Tarsometatarsal (Lisfranc) Joint

The tarsometatarsal (TMT) joint is the articulation between mid and forefoot segments and consists of the medial and lateral columns of the foot. The medial column is formed by the articulation of the medial, middle, and lateral cuneiforms with the proximal first, second, and third metatarsals, respectively (Fig 1). Within the medial column, the first cuneiform, metatarsal, and the first TMT articulation form the first ray. The lateral column is comprised of the cuboid, the proximal fourth and fifth metatarsals and the lateral aspect of the TMT (Figure 2). The TMT joint complex is structurally stabilized morphologically in a “Roman Arch” configuration and supported by extensive dorsal, plantar, and interosseous ligamentous network.²⁹ During locomotion, the degree in which the forefoot couples with the rearfoot varies dependent on velocity and the cardinal plane in which forefoot motion occurs.³⁰

First Ray & Hallux

The first ray, also referred to as the medial forefoot, is comprised of the first metatarsal, cuneiform, and the TMT joint. The first ray functions as a pillar and forms the distal truss of the medial longitudinal arch.³¹ The joint capsule and the dorsal tarsometatarsal, plantar tarsometatarsal, and plantar metatarsal ligaments provide passive stability to the TMT articulation. The hallux is comprised of the first proximal and distal phalanx and the interphalangeal joint. The medial collateral, lateral collateral, and plantar ligaments and the joint capsules provide stability to the first metatarsophalangeal and interphalangeal joints.

Lateral Forefoot

The lateral forefoot is comprised of the metatarsals and phalanges of the lateral four digits. The joint capsules and the medial collateral, and lateral collateral ligaments provide stability for each of the metatarsophalangeal and interphalangeal joints. A deep transverse and plantar metatarsal ligament supports

the intermetatarsal articulations. Together with the first ray, the metatarsals of the lateral forefoot form a metatarsal arch (Figure 3).

Extrinsic Control of the Ankle – Foot Complex

Traditionally, the extrinsic foot and ankle muscles are described as ‘prime movers’ based on the osteokinematic motion they cause when the foot is treated as a rigid segment. However, the authors aim to describe the functional roles of the extrinsic foot and ankle muscles in the context of a multisegmented foot and ankle complex. Collectively, the extrinsic and intrinsic foot and ankle musculature support, stabilize, dissipate force, and move the multiple articulations of the ankle and foot. Please refer to Figures 1-4 for illustrations of the extrinsic muscle tendons and their insertions.

LATERAL COMPARTMENT

The lateral compartment contains the fibularis longus and brevis, both of which are innervated by the superficial fibular nerve. The fibularis longus plays an important role in stabilization of the lateral midfoot and the first ray and is the primary evor of the forefoot. Originating in the lateral compartment, the tendon courses around a series of pulleys formed by the lateral malleolus, the peroneal tubercle, and the cuboid.^{27,32} The tendon exerts a compressive stabilizing force on the cuboid and contributes to midtarsal locking and prevention of midtarsal break.²⁷ The tendon continues from the cuboid in an anteromedial direction to the lateral plantar base of the first metatarsal. Functionally during stance, the fibularis longus stabilizes the medial column by everting the first ray and mechanically coupling the tarsometatarsal and naviculocuneiform joints through a torsion of the articular ligamentous and capsular tissues.³² The fibularis longus is the primary plantarflexor of the first ray^{32,33} and assists in the support of the medial longitudinal arch.³³ The fibularis longus contributes important afferent feedback in regards to ankle position. In a study investigating proprioception, balance, and reaction time in individuals who received a regional nerve block to the foot and ankle, it was concluded that the fibularis longus is a primary afferent input to the brain in maintaining balance even more so than the ligamentous structures of the

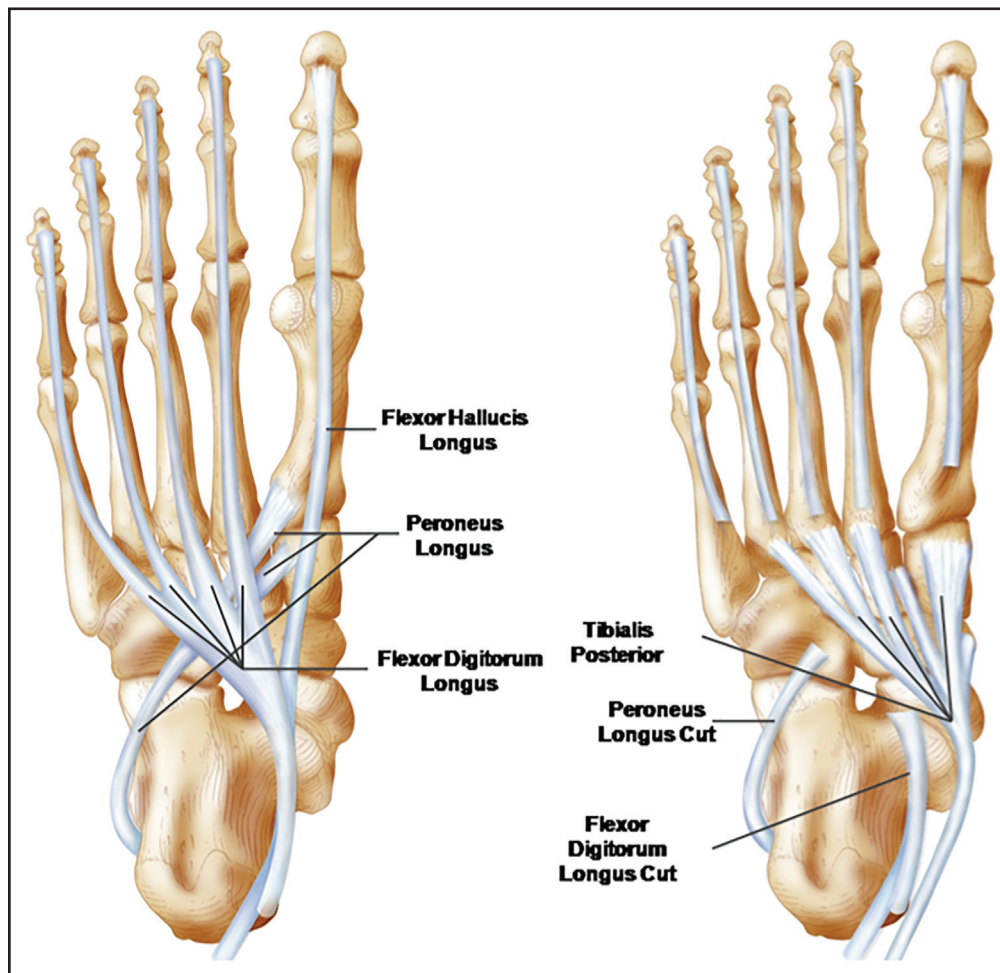


Figure 4. Extrinsic muscle tendons and insertions. Adapted from McKeon PO, Hertel J, Bramble D, Davis I. The foot core system: a new paradigm for understanding intrinsic foot muscle function. *Br. J. Sports Med.* 2014. Used with permission of the publisher.

ankle.³⁴ The fibularis longus, tibialis posterior, and the flexor digitorum longus tendons are oriented in a cross configuration and may contribute to the support and function of the longitudinal and transverse arches. Each tendon courses the span of both arches, with the fibularis longus traversing posterolateral to anteromedial and the flexor digitorum and posterior tibialis coursing posteromedial to anterolateral. It is plausible that synergistic concentric contraction of these muscles would contribute to approximation of opposing voussours and increase the rise of the longitudinal and transverse arches. It is also conceivable that like the long toe flexors, isometric or eccentric co-contraction of these muscles may contribute to stabilization and force attenuation of the foot arches.

The fibularis brevis originates in the lateral compartment of the leg and inserts on the styloid process

of the fifth metatarsal. The fibularis longus is the primary evolver of the forefoot and similarly, the fibularis brevis and fibularis tertius (of the anterior compartment) are the primary evolvers of the rear-foot and midfoot. In a study investigating the contribution of the fibularis longus and brevis to ankle and foot movement during a simulated early heel rise during the stance phase of gait, the fibularis brevis was found to be more effective in evolver the talonavicular and subtalar joints than the fibularis longus.³⁵

ANTERIOR COMPARTMENT

The fibularis tertius, tibialis anterior, extensor digitorum longus, and extensor hallucis longus originate in the anterior compartment and are innervated by the deep fibular nerve. The fibularis tertius, which

functions synergistically with the two lateral compartment muscles and was discussed in the previous section, inserts on the dorsal base of the fifth metatarsal. The tibialis anterior, extensor hallucis longus and the extensor digitorum longus course from the anterior compartment distally to the extensor retinaculum. The tibialis anterior tendon continues and inserts on the inferomedial aspect of medial cuneiform and the base of first metatarsal. During gait, the tibialis anterior is thought to have an important function in maintaining balance during the first quarter of stance.³⁶ The extensor hallucis and digitorum longus insert into the extensor apparatus of the great toe and the second through fifth toes, respectively.³⁷ The toe extensors, with the anterior tibialis, activate to lift the toes from the ground at terminal stance of gait.³⁸ They also assist in the stabilization of the talocrural and the tarsal bones during loading of the foot.³⁸ Afferent information provided from the muscles of the anterior compartment, in conjunction with lateral compartment muscles and cutaneous receptors, have been suggested to have an important role in ankle kinesthesia³⁹ and modulation of the soleus during the stance phase of gait.⁴⁰

SUPERFICIAL POSTERIOR COMPARTMENT

The gastrocnemius and soleus, collectively known as the triceps surae, innervated by the tibial nerve,

originate in the posterior compartment and insert on the calcaneal tuberosity. During stance phase, the triceps surae controls the forward progression of the shank on the talus^{38,41} and assists in providing a flexion moment at the knee.³⁸ Additionally, the triceps surae is able to mechanically influence the rearfoot, midfoot, and forefoot through its insertion on the calcaneus and ability to tension the plantar aponeurosis (PA), otherwise known as the plantar fascia (Figure 5). During gait, the PA is tensioned at 30% of stance until midstance, when elongation increases sharply.⁴² Peak elongation of the PA occurs at 80% of stance when the triceps surae exerts a plantarflexion moment on the calcaneus and the metatarsophalangeal joints of the toes are extended.⁴² The elongation of the PA elevates the medial longitudinal arch and inverts the midfoot and rearfoot via a shift in the calcaneal tendon medial to the axis of rotation of the subtalar joint.⁴³ Contraction of the triceps surae during weight bearing creates a plantarflexion moment in the rearfoot, tensions the PA, and increases the dorsiflexion moment and ground reaction force and in the midtarsal, cuneonavicular, and tarsometatarsal articulations.⁴⁴ An analysis using computer 3-D modeling of forefoot force transmission in the presence of impaired triceps surae strength found that contact area increased in the midfoot and plantar pressures

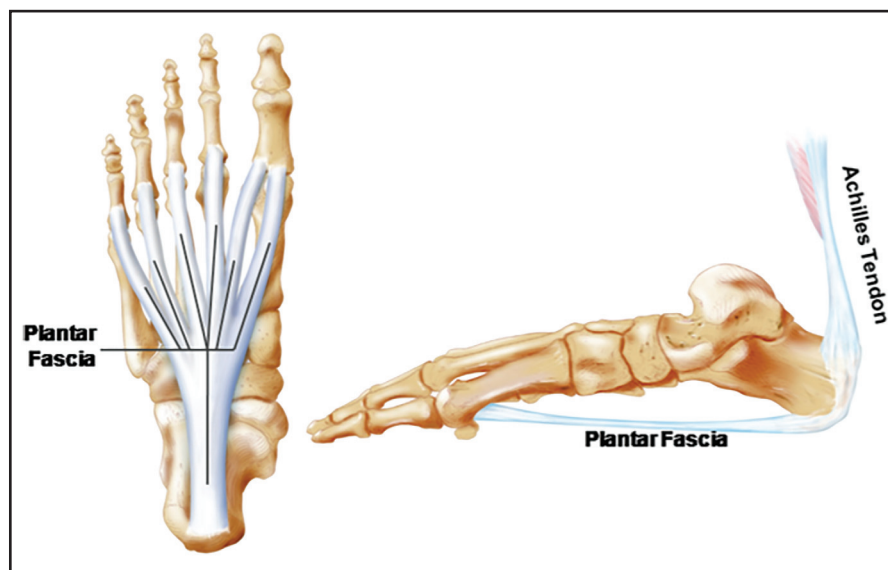


Figure 5. The windlass mechanism. The triceps surae forms a coupled relationship with the plantar fascia through the proximal attachment on the calcaneus and action on the rearfoot. Adapted from McKeon PO, Hertel J, Bramble D, Davis I. The foot core system: a new paradigm for understanding intrinsic foot muscle function. *Br. J. Sports Med.* 2014. Used with permission of the publisher.

decreased in the forefoot during terminal stance.⁴⁵ The triceps surae also has an important function as a proximal stabilizer of the calcaneus and in intrinsic foot muscle function. This was shown in patients with poliomyelitis who had impaired plantar flexor strength and unopposed intrinsic foot flexor function.⁴⁶ A foot deformity, coined the “calcaneus foot,” was a sequela of triceps surae strength impairment and was characterized by rearfoot dorsiflexion, forefoot plantarflexion, and clawing of the toes.⁴⁶

DEEP POSTERIOR COMPARTMENT

The flexor hallucis longus (FHL) and the flexor digitorum longus (FDL) tendons are innervated by the tibial nerve and traverse from the deep posterior compartment, through the flexor retinaculum, into the plantar aspect of the foot. The FHL is oriented longitudinally along the medial aspect of the foot. The FDL has a more oblique orientation from posteromedial to anterolateral. The FDL tendon is the proximal attachment for the lumbricals and the distal attachment for the quadratus plantae (flexor digitorum accessorius). While inserting distally and causing toe flexion, the FHL and FDL tendons also cross the transverse tarsal and TMT joints. During stance, the long toe flexors have been shown to contract isometrically to support and stabilize the longitudinal arches,^{33,47} provide afferent feedback,⁴⁷ and dissipate force during loading.^{47,48}

The tibialis posterior originates in the deep posterior compartment and courses deep to the flexor retinaculum.⁴⁹ It supports the medial longitudinal arch through its multiple insertions to the navicular tubercle, each tarsal bone, metatarsals two through four, and the flexor hallucis brevis muscle.⁴⁹ During barefoot walking, the tibialis posterior contracts to resist eversion and peroneal contraction and assists in stabilization of the rearfoot during initial contact (IC) and midstance.⁵⁰

Intrinsic Control of the Ankle – Foot Complex

Ten muscles in the plantar foot and two in the dorsal foot provide intrinsic control of the foot (Figure 6). The plantar intrinsic muscles are organized into four layers and are innervated by the medial and lateral plantar nerves, which branch from the tibial nerve. The dorsal intrinsic muscles are innervated by the

deep fibular nerve (extensor digitorum brevis) and the lateral plantar nerve (dorsal interossei muscles). During gait, the intrinsic foot muscles function both in open kinetic chain to shape the foot and toes in preparation for contact with the ground and in closed kinetic chain to accommodate the terrain and during force attenuation and transmission.

FIRST PLANTAR LAYER

The superficial layer is comprised of the abductor hallucis, flexor digitorum brevis, and the abductor digiti minimi. The abductor hallucis inserts proximally on the medial tubercle of the calcaneus, courses proximal to distal on the medial foot, and inserts distally on the medial base of the first proximal phalanx. The flexor digitorum brevis inserts proximally on the medial tubercle of the calcaneus, courses proximal to distal in the center of the plantar foot, and inserts distally on the middle phalanx of digits 2-5. The abductor digiti minimi inserts proximally on the medial and lateral tubercles of the calcaneus, courses proximal to distal on the lateral foot, and inserts distally on the lateral base of the fifth proximal phalanx. These muscles abduct the hallux, flex toes 2-5, and abduct the little toe when functioning in open kinetic chain, respectfully. During the stance phase of gait, the orientation and location of the abductor hallucis and abductor digiti minimi in relation to the medial and lateral longitudinal arches position these muscles to contribute to stabilization and eccentric control of arch descent during loading. The flexor digitorum brevis is analogous to the flexor digitorum profundus in the upper extremity based on morphology, insertion, and relationship to the long digit flexor. While speculative, it is likely that the flexor digitorum brevis contributes to eccentric control of metatarsophalangeal extension from mid-stance to pre-swing phase of gait.

SECOND PLANTAR LAYER

The second layer is comprised of the quadratus plantae (flexor digitorum accessorius) and the lumbrical muscles, all of which attach to and function with the flexor digitorum longus. The quadratus plantae originates on the plantar calcaneus and courses proximal to distal to insert on the flexor digitorum longus tendon. The four lumbrical muscles originate on the flexor digitorum longus tendon proximally and course



Figure 6. Intrinsic muscles of the foot. Plantar intrinsics: Layer 1: 1 = abductor hallucis, 2 = flexor digitorum brevis, 3 = abductor digiti minimi; Layer 2: 4 = quadratus plantae, 5 = lumbricals 1-4; Layer 3: 6 = flexor digiti minimi, 7a = adductor hallucis oblique head, 7b = adductor hallucis transverse head, 8 = flexor hallucis brevis; Layer 4: Dorsal Interossei. Dorsal Intrinsics: 10 = dorsal interossei, 11 = extensor digitorum brevis. Adapted from McKeon PO, Hertel J, Bramble D, Davis I. The foot core system: a new paradigm for understanding intrinsic foot muscle function. *Br. J. Sports Med.* 2014. Used with permission of the publisher.

distally to insert on the digital extensor expansions. In open kinetic chain, the quadratus plantae assists in long toe flexion and the lumbricals perform metatarsophalangeal flexion and interphalangeal extension. The proximodistal orientation of the quadratus plantae in relationship to the flexor digitorum longus allows for a change in force vectoring of the long toe flexor tendon from a proximomedial-distolateral angle to a more longitudinal vector. During gait, the lumbricals likely contribute to eccentric control of metatarsophalangeal extension while extending the interphalangeal joints during toe-break. It is also plausible that the lumbricals function by pulling the

flexor digitorum longus tendons distally when the ankle is plantarflexed to optimize length-tension and prevent active insufficiency during toe flexion.

THIRD PLANTAR LAYER

The third plantar layer is comprised of the flexor digiti minimi, the adductor hallucis, and the flexor hallucis brevis. The flexor digiti minimi inserts proximally on the base of the fifth metatarsal and travels distally to insert on the base of the fifth proximal phalanx. The adductor hallucis has two heads, an oblique and transverse head. The oblique head inserts proximally on the base of the second through

fourth metatarsals and traverses distal medial to insert on the base of the first proximal phalanx. The transverse head inserts laterally on the plantar metatarsophalangeal ligaments and travels medially to insert with the oblique head on the base of the first proximal phalanx. The flexor hallucis brevis inserts proximally on the cuboid, lateral cuneiform, and the middle and medial cuneiform by way of the tibialis posterior tendon and travels distally to insert on the base of the first proximal phalanx. In open kinetic chain, these muscles flex the little toe, adduct the hallux, and flex the hallux, respectively. During the stance phase of gait, the flexor hallucis brevis and the flexor digiti minimi likely contribute to eccentric control of toe extension from midstance to pre-swing. The adductor hallucis likely contributes to stabilization of the first ray during forefoot loading in the latter half of stance.

FOURTH PLANTAR LAYER

The fourth and deepest plantar layer is comprised of the three plantar interossei. The interossei insert proximally on the medial shafts of metatarsals 3-5 and traverses distally to insert on the bases of the proximal phalanges. These muscles adduct the toes in open kinetic chain and likely provide isometric or eccentric control of toe splay during forefoot loading in pre-swing phase of gait. It has also been speculated that these muscles stabilize the tarsometatarsal joints in conjunction with the dorsal interossei during late stance phase of gait.⁵¹

Dorsal Intrinsic Muscles

The two intrinsic muscles on the dorsal foot are the extensor digitorum brevis and the four dorsal interossei muscles. The medial head of the extensor digitorum brevis is sometimes treated as a separate muscle, the extensor hallucis brevis. For the purpose of this manuscript, the extensor hallucis brevis will be treated as integral with the extensor digitorum brevis. The extensor digitorum brevis inserts proximally on the calcaneus and courses distally to insert on the tendons of the extensor digitorum longus (toes 2-4) and the base of the first proximal phalanx. The dorsal interossei insert proximally on the shafts of the metatarsals one through five and traverse distally to insert on the proximal phalanges. In open kinetic chain, these muscles abduct and extend toes

two through four, respectfully. The dorsal interossei also function with the palmar interossei to stabilize the forefoot in pre-swing phase of gait.⁵¹

In the sentinel study performed by Mann and Inman,⁵² the intrinsic foot muscles were found to work synergistically as a functional unit during gait to provide stabilization of the midfoot, and that greater muscular activity was required to stabilize the foot in individuals who had excessive pronation observed during static standing. More recently, McKeon and colleagues⁸ described a “foot core” system that is analogous to the lumbopelvic complex and comprised of active, passive, and neural subsystems. The intrinsic muscles play an important direct role in both active and neural subsystems and indirectly to the passive subsystem.⁸ The flexor digitorum brevis, flexor hallucis brevis, oblique head of the adductor hallucis, and abductor digiti minimi are orientated longitudinally and run perpendicular to the transverse tarsal joint surface, making them prime stabilizers for this articulation and for the longitudinal arch.⁵² The intrinsic muscles are stretched with deformation of the medial longitudinal arch during loading.²⁵ It is reasonable to assume that the stretch of the musculotendinous sensory organs provide afferent feedback during foot deformation, shaping, and force attenuation. Conversely, concentric contraction of the plantar intrinsic foot muscles produces calcaneal and metatarsal displacement resulting in decreased arch length and increased arch height.²⁵ This alteration of medial longitudinal arch morphology during IFM contraction forms the basis of the short foot exercise, an intervention utilized clinically to strengthen the foot core in the treatment of ankle-foot impairment.^{8,9} It is thought that the intrinsic muscles, when functioning in conjunction with active extrinsic muscle contraction and the PA, contribute to buttressing of the foot during force attenuation and transmission.²⁵ Electromyographic analysis of muscle function during gait demonstrated coordinated activation of extrinsic and intrinsic toe flexors activity in the mid to terminal stance, sequentially followed by extrinsic and intrinsic toe extension activity in terminal stance to early swing phase.⁵³ The coordinated extrinsic and intrinsic activity observed in this study provides further evidence of the role of the intrinsic foot muscles during force attenuation and transmission in gait.

MULTISEGMENTED ANKLE-FOOT COMPLEX KINEMATICS DURING GAIT

Rearfoot

When referenced to the shank, the rearfoot is in a neutral sagittal, pronated, and abducted position at IC and transitions to a plantarflexed, pronated, and adducted position during early stance phase.⁷ During midstance, the rearfoot is dorsiflexed, pronated, and abducted until 70% of stance when the rearfoot becomes plantarflexed and supinated at 90% of stance.⁷ Total magnitude of rearfoot excursion is 10-15° in all three planes.⁷

Midfoot

When referenced to the rearfoot, the midfoot is supinated at IC and moves to a dorsiflexed, pronated, and adducted position post IC.⁷ From 15% to 80% of stance, the midfoot is maintained in neutral in the sagittal plane, pronation, and abduction. The midfoot is plantarflexed, supinated, and continues into abduction in the last 20% of stance.⁷ Total magnitude of midfoot range of motion is 5-8° in all three planes.⁷

Mann⁵⁴ described a coupling of the rearfoot and forefoot by way of the midfoot during mid to terminal stance when shank external rotation and rearfoot supination causes the longitudinal axes of talonavicular and calcaneocuboid to diverge, creating a more rigid lever for push off. Position and control of the medial longitudinal arch has been found to be contributory to rearfoot to forefoot coupling.^{55,56} The transverse tarsal joint has been described to have two different modes of function which are dependent on mechanical demands of walking and running at various speeds, with different loads, on various surfaces, and whether the cuboid is locked by the fibularis longus.²¹ The individual axes of the transverse tarsal joint have a corresponding parallel axis at the metatarsophalangeal joints.²¹ During pushoff, the lever arm of the foot changes dependent on which axis the foot is functioning about.²¹ In “low gear” dominated activity such as uphill walking with a load or the first steps of a sprint, pushoff occurs with the rearfoot adducted and plantarflexed while cuboid rotation occurs about the oblique axis in the transverse tarsal joint.²⁶ Plantar pressure progression is shifted to the lateral forefoot and results in a

functionally shortened lever arm as toe break occurs about the axis formed by the lesser metatarsophalangeal joints.^{21,26,57} With the foot adducted, ground contact during low gear push off is transmitted from the lateral heel to the lateral aspect of the first metatarsal head.²⁶ During pushoff in “high gear” dominated activity such as fast level walking and sprinting, the plantar aponeurosis (PA) is tensioned and the fibularis longus compresses and everts the calcaneocuboid joint to a closed pack position, mechanically coupling the rearfoot to forefoot to prevent midtarsal break.²⁶ With the foot neutral in the transverse plane, force is transmitted through the medial border of the first metatarsal head and hallux as push off occurs about the transverse metatarsophalangeal axis.²⁶ The ability of the foot to pushoff about two axes may contribute to balance and negotiation of uneven ground by allowing for alteration of forward progression in response to perturbation.⁵⁷ More recent kinematic studies using multisegmented foot models have found that individual foot segments remain compliant during ambulation.⁵⁸⁻⁶⁰ During gait, forefoot inversion and eversion motion remains relatively uncoupled and independent from frontal plane motion in the rearfoot.⁶⁰ However, a coupling relationship has been found between rearfoot pronation and forefoot dorsiflexion in the first half of stance and forefoot plantarflexion and rearfoot supination in the latter half of stance, with vector coded mean excursions of 41-43°.⁶⁰ Additionally, there is a high degree of coupling (mean excursions 24-32°) between rearfoot frontal plane motion and forefoot transverse motion during walking in the first 20% of stance and again from midstance to preswing.⁶⁰

Lateral Forefoot

The lateral forefoot is plantarflexed, slightly supinated, and abducted at IC, followed by plantarflexion, pronation, and abduction post IC relative to the midfoot.⁷ Lateral forefoot pronation occurs in the latter half of IC until the forefoot contacts the ground.⁷ The metatarsal arch deforms and widens in a metatarsal forming phase, with loading incurred in the lateral aspect of the forefoot.⁶¹ From 15% to 70% of stance, the lateral forefoot remains dorsiflexed while slightly supinated.⁷ Force distribution across the forefoot becomes more uniform during mid stance as the medial forefoot is progressively

loaded.⁶¹ From 70% to 90% of stance, the lateral forefoot supinates⁷ and the metatarsal arch increases to its maximal height⁶¹. Despite direct loading of the metatarsal arch in the last 25% of stance, forefoot width decreases and tightens as the foot prepares to pushoff.⁶¹ It is speculated that the paradoxical increase in metatarsal arch height during maximal loading is resultant from the engagement of connective tissues (such as the PA) and contractile support mechanisms of the arch.⁶¹ Total lateral forefoot excursion is approximately 10° in each plane.⁷

There are two axes of rotation formed by the metatarsal heads that parallel the axes of transverse tarsal joint.^{21,26,57,62,63} The axis that the body employs during propulsion is dependent on requirements of force generation or balance requirements and is dictated by midfoot function.^{21,26,57,62,63} The three major functions for the metatarsal heads include adaptation to changing gravitational axes in balance, transfer of weight from rearfoot to the forefoot prior to terminal stance, and function as a lever for propulsion in terminal stance.⁶² Of these functions, the role of the metatarsal heads in providing fine adjusting movements of the foot has been described as the most important in maintaining balance during standing and walking.⁶² Impaired ability of the metatarsals to perform fine adjusting movements would shift the burden of balance to the subtalar joint and result in maximal impairment,⁶² a scenario likely to be observed clinically when tarsometatarsal hypomobility or impaired neuromuscular function is a consequence post injury. During ambulation that occurs at lower velocity, is balance intensive, involves negotiating inclines, or includes carrying loads, the foot is adducted so pushoff may occur about the oblique axis that transects metatarsal heads 2-5 in the lateral forefoot.²¹

First Ray & Hallux (Medial Forefoot)

In relation to the midfoot, the first ray dorsiflexes, everts, and adducts at IC, abducts during early stance phase post IC, and remains dorsiflexed and everted until 75% of stance.⁷ During the last quarter of stance, the first ray dorsiflexes, inverts, and adducts.⁷ Total excursion in the first ray is 6-16° in all planes.⁷ When referenced to the first ray, the hallux is dorsiflexed, supinated, and abducted at IC followed by slight pronation at 15% of stance that persists until

the last quarter of stance.⁷ The hallux dorsiflexes and abducts during the last 25% of stance.⁷ Total hallux excursion is 55°, 50°, and 18° in the sagittal, transverse, and frontal planes, respectively.⁷

During higher velocity ambulation, running, and sprinting, the foot is neutral in the transverse plane so that pushoff may occur about the transverse metatarsal axis.^{21,26} The transverse axis transects the metatarsal heads of digits 1 and 2 and parallels the transverse axis of the midtarsal joint.^{21,26,57,62,63} When the forefoot is coupled to the rearfoot and the transverse metatarsal axis employed during pushoff, the moment arm is functionally lengthened by 20%.²¹

CONCLUSIONS

In summary, the multiple segments that comprise the ankle-foot complex function synergistically to transmit and attenuate force during propulsion, accommodate and conform with uneven terrain, and provide important afferent information and motor adjustment to maintain balance during standing and ambulation. The interplay of segmental joint mobility, sensorimotor function, and movement and stabilization provided by the extrinsic and intrinsic musculature is required to coordinate and execute the complex kinematics observed in the ankle-foot complex during propulsion. In part two of this clinical commentary, alterations in kinematics secondary to joint and neuromotor impairment incurred following lateral ankle sprain and in chronic ankle instability will be discussed.¹⁰ Clinical assessment and treatment techniques for the ankle-foot complex will also be addressed.¹⁰

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